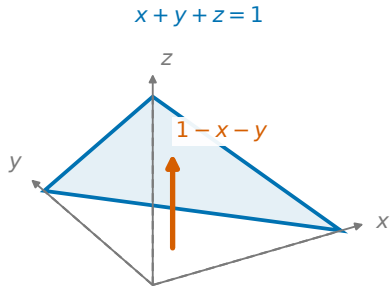
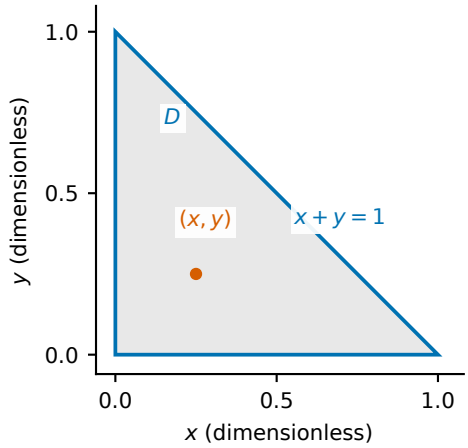
A



$0 \le z \le 1 - x - y$

B



$$V = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz\, dy\, dx = 1/6$$